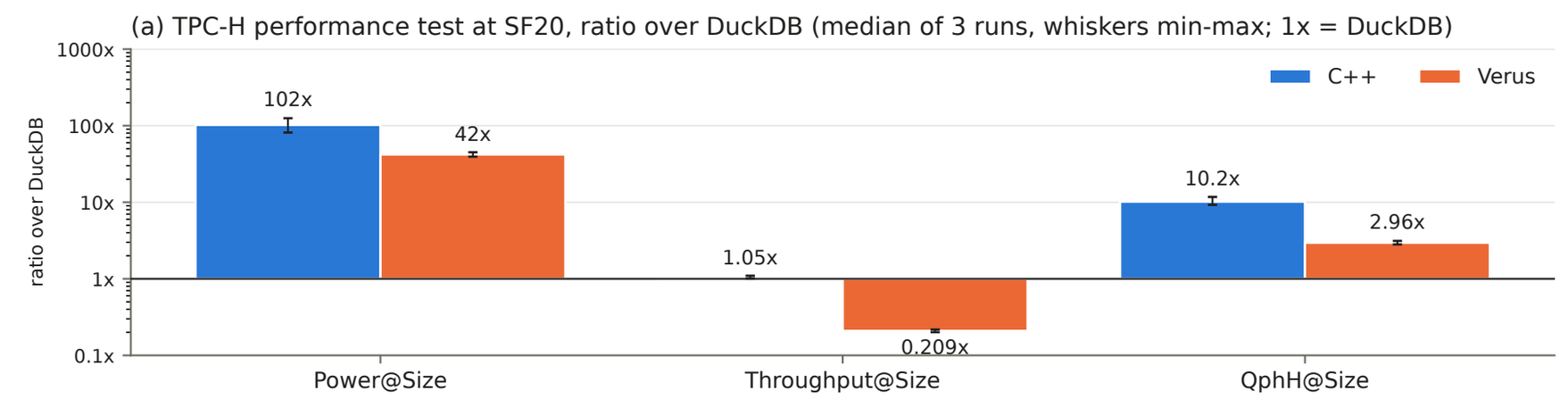
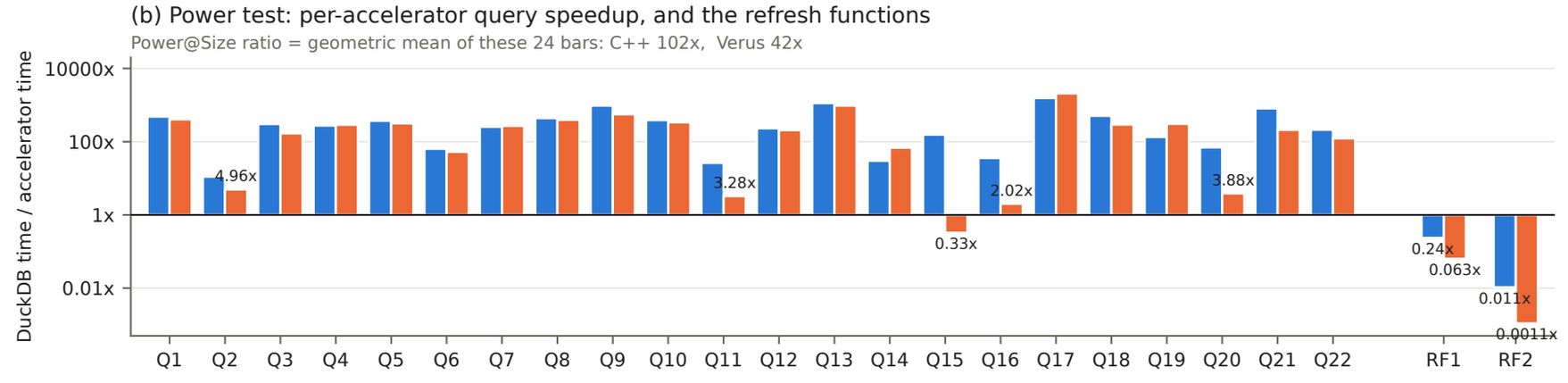
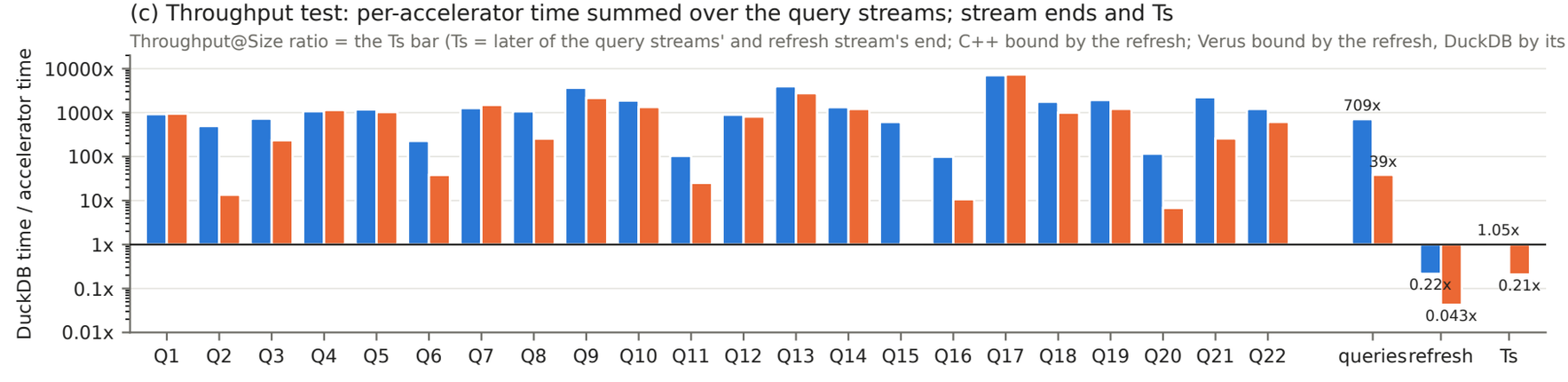
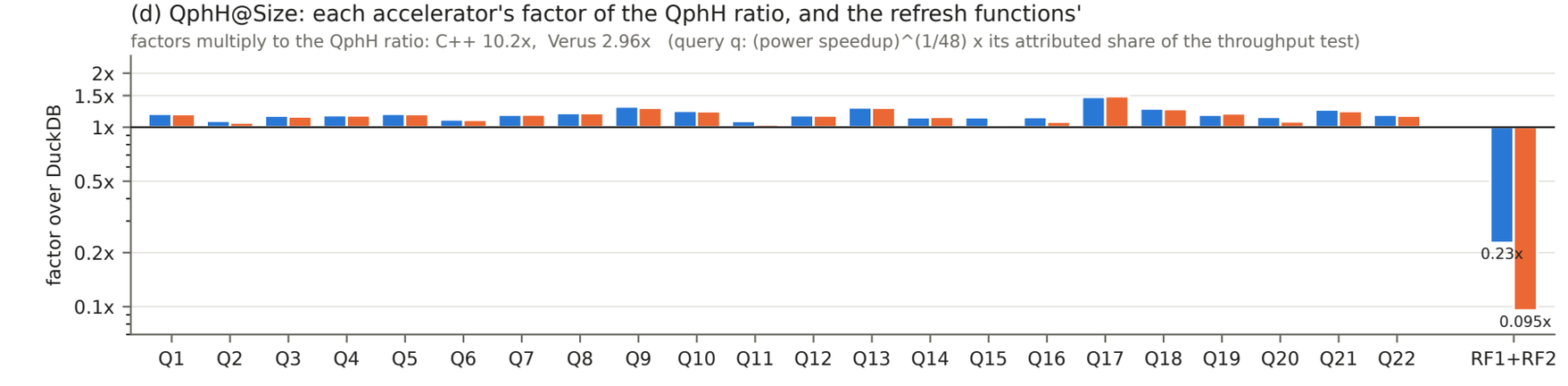

(a) TPC-H performance test at SF20, ratio over DuckDB (median of 3 runs, whiskers min-max; 1x = DuckDB)
ratio over DuckDB
1000x
100x
10x
1x
0.1x
C++
Verus
102x
42x
1.05x
0.209x
10.2x
2.96x
Power@Size
Throughput@Size
QphH@Size


(b) Power test: per-accelerator query speedup, and the refresh functions
Power@Size ratio = geometric mean of these 24 bars: C++ 102x, Verus 42x
DuckDB time / accelerator time
10000x
100x
1x
0.01x
4.96x
3.28x
0.33x
2.02x
3.88x
0.24x
0.063x
0.011x
0.0011x
Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12 Q13 Q14 Q15 Q16 Q17 Q18 Q19 Q20 Q21 Q22 RF1 RF2


(c) Throughput test: per-accelerator time summed over the query streams; stream ends and Ts
Throughput@Size ratio = the Ts bar (Ts = later of the query streams' and refresh stream's end; C++ bound by the refresh; Verus bound by the refresh, DuckDB by its queries)
DuckDB time / accelerator time
10000x
1000x
100x
10x
1x
0.1x
0.01x
709x
39x
0.22x
0.043x
1.05x
0.21x
Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12 Q13 Q14 Q15 Q16 Q17 Q18 Q19 Q20 Q21 Q22 queries refresh Ts


(d) QphH@Size: each accelerator's factor of the QphH ratio, and the refresh functions'
factors multiply to the QphH ratio: C++ 10.2x, Verus 2.96x (query q: (power speedup)^(1/48) x its attributed share of the throughput test)
factor over DuckDB
2x
1.5x
1x
0.5x
0.2x
0.1x
0.23x
0.095x
Q1 Q2 Q3 Q4 Q5 Q6 Q7 Q8 Q9 Q10 Q11 Q12 Q13 Q14 Q15 Q16 Q17 Q18 Q19 Q20 Q21 Q22 RF1+RF2